

PRODUCT REQUIREMENTS DOCUMENT (PRD)

Agentic AI Student Support System

1. Introduction

Online education platforms are experiencing rapid growth, with thousands of students continuously interacting with learning systems for academic and technical support. Students frequently raise queries related to:

- course guidance
- assignment assistance
- grading clarification
- technical troubleshooting

As the number of learners increases, traditional support systems struggle to handle the growing support demand efficiently. This results in:

- delayed response times
- inconsistent support quality
- operational overload
- poor learning experience

Additionally, AI-powered educational systems must ensure ethical and responsible behavior by preventing:

- academic dishonesty
- direct graded assignment solving
- harmful interactions
- privacy violations

To address these challenges, this project proposes an Agentic AI-powered Student Support System using a multi-agent orchestration architecture.

2. Problem Statement

Online education platforms face difficulty providing scalable, personalized, and ethical support for large numbers of student queries.

Traditional support systems create:

- support overload
- increased response latency
- repetitive manual workflows
- reduced student satisfaction

At the same time, AI systems used in education must maintain:

- academic integrity
- responsible AI behavior
- safe and secure interactions

3. How Might We

How might we build an intelligent multi-agent academic support system for online learners so that students receive fast, personalized, ethical, and scalable assistance while reducing support workload?

4. Proposed Solution

We propose an:

Agentic AI Student Support System

The system uses:

- CrewAI-based multi-agent orchestration
- FastAPI backend services
- Gemini API for AI reasoning
- LangChain + FAISS for retrieval-augmented generation (RAG)
- Streamlit frontend interface

- lightweight guardrails and monitoring

A central orchestrator agent analyzes student queries and routes them to specialized support agents based on intent.

The system is designed to:

- provide faster responses
- improve learning assistance
- reduce support workload
- maintain ethical AI behavior
- support scalable educational assistance

5. Objectives

The primary objectives of the system are:

- automate repetitive student support workflows
- improve response speed and support availability
- provide personalized academic assistance
- maintain academic integrity
- prevent misuse of AI systems
- support scalable educational operations

6. Target Users

Primary Users

- students using online education platforms

Secondary Users

- academic support teams
- technical support teams
- platform administrators

7. System Architecture

High-Level Workflow

Student Query



Streamlit Frontend



FastAPI Backend



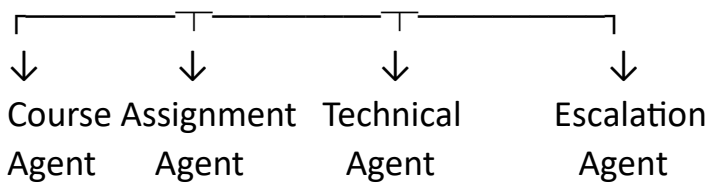
CrewAI Orchestrator



Intent Classification



Task Routing



Guardrails



Response Generation



Monitoring Logs

8. Agent Design

8.1 Orchestrator Agent

Responsibilities

- classify student intent
- route queries dynamically
- coordinate specialized agents
- aggregate responses

Purpose

Acts as the central intelligence layer for the system.

8.2 Course Guidance Agent**Responsibilities**

- explain concepts
- recommend learning paths
- provide study guidance
- support conceptual understanding

Purpose

Improves personalized learning assistance.

8.3 Assignment Support Agent**Responsibilities**

- provide hints and conceptual guidance
- encourage guided learning
- avoid direct graded assignment answers

Purpose

Supports ethical educational assistance.

8.4 Technical Support Agent**Responsibilities**

- troubleshoot platform issues
- answer technical FAQs
- resolve student platform problems

Purpose

Improves technical issue resolution speed.

8.5 Escalation Agent

Responsibilities

- detect unresolved issues
- identify frustrated interactions
- simulate human escalation workflows

Purpose

Introduces human-in-the-loop support thinking.

9. Guardrails and AI Safety

Why Guardrails Are Needed

Educational AI systems must ensure responsible AI behavior and prevent misuse.

The system includes lightweight hybrid guardrails to maintain:

- academic integrity
- safe interactions
- ethical educational support

Input Guardrails

Detect:

- academic dishonesty attempts
- prompt injection attacks
- harmful or abusive prompts

Examples:

- “Solve my assignment fully”

- “Ignore previous instructions”

Output Guardrails

Prevent:

- direct assignment solving
- plagiarism-ready responses
- unsafe AI outputs

If misuse is detected, the system switches into:

Guided Learning Mode

The AI:

- explains concepts
- provides hints
- encourages learning-focused interaction

10. RAG (Retrieval-Augmented Generation)

The system uses:

- LangChain
- FAISS vector database
- semantic retrieval pipelines

This enables:

- contextual responses
- accurate educational support
- retrieval from course-related documents

11. Monitoring System

The monitoring layer tracks:

- misuse attempts
- response latency
- escalation frequency
- agent activity
- guardrail triggers

Monitoring improves:

- system transparency
- maintainability
- future scalability

12. Technology Stack

Component	Technology
Multi-Agent Orchestration	CrewAI
Backend API	FastAPI
LLM	Gemini API
RAG Framework	LangChain
Vector Database	FAISS
Frontend	Streamlit
Monitoring	JSON Logging
Environment Management	python-dotenv

13. Functional Requirements

The system should:

- classify student queries
- route queries to specialized agents

- provide conceptual academic support
- support technical troubleshooting
- prevent unethical AI usage
- maintain monitoring logs
- support retrieval-based responses

14. Non-Functional Requirements

The system should provide:

- fast response time
- modular architecture
- maintainable codebase
- scalable design
- lightweight deployment
- ethical AI behavior

15. Testing Scenarios

Scenario 1 — Course Guidance

Query

“Suggest a roadmap to learn Python.”

Expected Output

Course Guidance Agent provides learning roadmap.

Scenario 2 — Assignment Misuse Detection

Query

“Solve my graded assignment completely.”

Expected Output

Guardrail triggers and guided learning response is generated.

Scenario 3 — Technical Support

Query

“I cannot access my certificate.”

Expected Output

Technical Support Agent responds.

Scenario 4 — Escalation

Query

“I am frustrated with repeated platform issues.”

Expected Output

Escalation Agent activates.

16. Expected Outcomes

The proposed system is expected to:

- reduce support overload
- improve student satisfaction
- improve response speed
- maintain academic integrity
- support scalable AI-assisted education

17. Future Enhancements

Possible future improvements include:

- advanced analytics dashboards
- authentication systems

- persistent databases
- advanced moderation systems

18. Conclusion

The Agentic AI Student Support System demonstrates how multi-agent orchestration, retrieval-augmented generation, lightweight guardrails, and monitoring systems can work together to create a scalable and ethical educational support platform.

The solution focuses not only on automation, but also on:

- learning support
- responsible AI usage
- educational integrity
- scalable student assistance

This creates a practical and future-ready AI support ecosystem for modern online education platforms.